Postpartum Physiological Changes

Goal:

To provide students with a comprehensive understanding of the normal physiological changes that occur in the postpartum period and to equip them with the knowledge and skills necessary to assess, monitor, and manage these changes effectively to promote maternal health and well-being.

Objectives:

At the end of this lecture, the student will be able to:

- ☞ Describe the normal physiological changes that occur during the postpartum period.
- ☞ Differentiate between the normal and abnormal physiological changes that occur during the postpartum period.
- ☞ Discuss nursing care during postpartum period.
- ☞ Use critical thinking exercise to improve selecting nursing care plane.

Puerperium:

The first 6 weeks "40 days" following the birth of an infant in which maternal body in general and genital organs in particular return to nearly pre-pregnancy state.

Stages of Postpartum

The postpartum period is generally divided into **three main stages**:

1. Immediate Postpartum (First 24 hours)

- Begins **right after delivery** (whether vaginal or cesarean).
- **Critical period** for monitoring complications such as:

- Postpartum hemorrhage (PPH),
- Uterine atony
- Preeclampsia/eclampsia
- Infection
- Thromboembolism.

- **Key assessments:**

- Vital signs,
- Fundal height and tone,
- Vaginal bleeding (lochia),
- Bladder function,
- Pain management.

2. Early Postpartum (Day 2 to Day 7)

- Ongoing **physical recovery** and **bonding**.
- Uterus continues to involute.
- **Lochia** transitions:
  - **Lochia rubra** (bright red, lasts 3–5 days),
  - Breast milk begins to be established,
  - Psychological monitoring (baby blues, risk of postpartum depression begins).
- Encourage **ambulation**, perineal hygiene, breastfeeding support.

3. Late Postpartum (Week 2 to 6 or up to 12 weeks)

- Full recovery of the reproductive organs and return of menstruation (if not breastfeeding).
- Monitoring for:
- Postpartum depression or anxiety,
- Delayed hemorrhage or infection,
- Contraceptive counseling,
- Pelvic floor dysfunction or urinary incontinence.
- Recommended postnatal check-up: 4–6 weeks after delivery.

Post-Partum Physiological Changes:

1} Reproductive system:

Involution of the uterus: -

It refers to changes that reproductive organ, especially the uterus, undergo after childbirth to return to their nearly pre-pregnancy size and condition.

The involution occurs by two processes:

a. Autolysis "self-digestion": The protein material of the muscle fibers is broken down by certain enzymes —proteolytic enzyme ‖and absorbed in the blood stream and excreted by the kidneys in the urine.

b. Ischemia "Decreased blood supply"

- ☞ Contraction and retraction of the uterine muscle fibers compresses. The blood vessels and reduces the blood supply to the uterus.
- ☞ The old blood vessels become obliterated by thrombosis, and then undergo degenerative changes.
- ☞ The remains of blood vessels can be detected as elastic fibers in the multiparous uterus.
- ☞ Endometrial regeneration is completed by about 3 weeks except for the placental site, where regeneration is not complete until 6 weeks.

Involution rate:

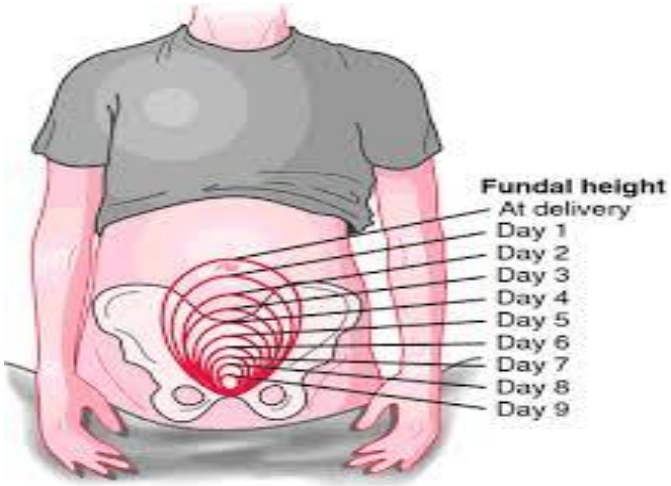
- Immediately following delivery, the size of uterus as large grapefruit and can be palpated midway between the symphysis pubis and umbilicus,
- Within an hour, the fundus rises to the level above the umbilicus and should remain at this level for about 24 hours the uterus now weighs approximately 1000 gm.
- After 24 hours, the fundus begins to descend by approximately 1cm, or one fingerbreadth, per day, so that by the 10th day it is in the pelvic cavity and cannot be palpated abdominally.
- Within a week, the weight of the uterus decreases to about 500g; at 6 weeks, the uterus weight 60g.This is roughly the pre-pregnancy weight.

Monitor uterus and vaginal bleeding,

- Every 30 minutes for first hour Postpartum.
- Then hourly for 2 more hours, every 4 hours.
- Every 8 hours or more frequently if:
- There is bogginess.
- Position out of midline.
- Heavy lochia flow.

Determine firmness of fundus and as certain position approximate descent of 1 cm or 1 fingerbreadth per day.

If boggy (soft), gently massage top of uterus until firm – notify health care provider if does not firm.



Factors Affecting Fundal Level Postpartum:

1. Bladder Fullness
- A **distended bladder** pushes the uterus upward and to the side (usually right), which can interfere with uterine involution and increase the risk of postpartum hemorrhage.

2. Uterine Atony
- Lack of uterine muscle tone causes the uterus to remain soft and enlarged, leading to a higher-than-expected fundal height.

3. Multiparity
- Women who have had multiple pregnancies may experience slower uterine involution, causing a slightly higher fundal level than in primiparas.

4. Retained Placental Fragments
- Retained tissue interferes with normal uterine contraction and involution, delaying fundal descent.

5. Infection (Endometritis)
- Postpartum uterine infection can cause subinvolution and tenderness, resulting in a higher fundal position.

6. Multiple Gestation or Polyhydramnios

- Overdistention of the uterus during pregnancy can lead to slower uterine shrinkage postpartum.

7. Breastfeeding

- Stimulates oxytocin release, which promotes uterine contraction and faster fundal descent.

8. Cesarean Delivery

- May result in slightly delayed uterine involution, depending on surgical and recovery factors.

9. Uterine Fibroids

- Can alter the shape and descent of the uterus, potentially keeping the fundus at a higher level.

10. Postpartum Hemorrhage

- Causes uterine distention and may lead to a boggy and elevated fundus.

Lochia:

✓ It is the uterine discharge coming through the vagina during the first 3-4 weeks or the postpartum.

✓ It is alkaline in reaction; the amount is more than the menstrual flow, with fleshy odor.

✓ It contains blood, fibrin, leucocytes, dead decidual tissue, vaginal cells, peptone, cholesterol, and numerous nonpathogenic bacteria.

Types of Lochia:

1-Lochia rubra (1-3 days):

is the first red in color discharge, composed of blood, shreds of fetal membranes, decidua, vernix caseosa, lanugo and membranes. It is red in color because of the large amount of blood it contains. It lasts 1to 4 days after birth, sometimes to7days.

2-Lochia Serosa (4-10 days):

Pink, yellow discharge.

It contains serous exudates, erythrocytes, leukocytes, cervical mucus and microorganisms.

3-Lochia Alba (or purulent) (11-17 days):

Creamy or white discharge**.**

It contain leukocytes and mucus

Cervix:

☞ Small tears or lacerations may be present, and the cervix is often edematous.

☞ Rapid healing takes place, and the end of the first week the cervix feels firm and the external so is the width of a pencil.

☞ The internal os closes as nearly before pregnancy, but the shape of the external os is permanently changed.

☞ It remains slightly open and appears slit-like rather than round, as in the nulliparous women.

Vagina:

- The vagina diminishes in size, but not as the pre gravida state. Rugea reappears in the third week. These are small skin folds in the lower part, dark red in color.
- The vagina and vaginal introitus are greatly stretched during birth to allow passage of the fetus. Soon after childbirth, the vaginal walls appear edematous, and multiple small lacerations may be present.
- The hymen is permanently torn and heals with small, irregular tag of tissue visible at the vaginal introitus.

Vulva:

- Edema, minute or frank laceration may be seen immediately after labor. Edema disappears gradually in a few days while lacerations, if not properly mended by sutures, may lead to the formation of a postpartum ulcer.

Perineum:

- The perineum has been stretched and thinned to accommodate the size of the infant.
- The pelvic floor muscles are overstretched and weak. The appearance of the perineum will vary greatly, depending on the type and extent of the episiotomy or laceration.
- Commonly the soft tissues of the perineum are edematous and bruised.
- Ecchymosis due to rupture of surface capillaries.

Resumption of Ovulation and Menstruation:

- Most mothers resume menstruation within 7 to 9 weeks after childbirth.
- In lactating mothers, menstruation usually 3-4 months, and sometimes as late as 24 months.
- The first period is generally profuse and prolonged.
- It should be mentioned that ovulation can commence in the absence of menstruation, and another pregnancy can occur

Fertility Symptoms after Childbirth

Tips to ensure effective lactational amenorrhea contraceptive method:

Ovulation Symptoms

Body weight: Loss of weight is observed during the first l0 days particularly in the non-lactating mothers. There is about a 4 – 5 kg loss of body weight (sometimes 8 kg) due to evacuation of uterine contents and diuresis.

2} Cardiovascular System:

Hemodynamic changes in pregnancy and postpartum

Include a comparison between the following parameters; CO cardiac output, HR heart rate, RBC red blood cell, SBP systolic blood pressure, SVR systemic vascular resistance

Blood volume:

Following delivery, despite 300 to 500 ml of blood loss during normal vaginal delivery, and 500-1000 ml is lost in cesarean births, excess blood volume, which was necessary during pregnancy, remains in the intravascular compartment and in interstitial spaces.

The body rids itself of the excess fluids by two methods.

Diuresis: "increased excretion of urine" is facilitated by a decline in the adrenal hormone aldosterone, which is increased during pregnancy to counteract the salt-wasting effect of progesterone.

As aldosterone production decreases, sodium retention declines and fluid excretion accelerate. A urinary output of 3000 ml per day is not common for the first few days of the postpartum period.

Diaphoresis "profuse perspiration" also rids the body of excess fluids through skin" sweating often occurs at night" can be uncomfortable for the mothers who is not prepared for it, explanation of the cause and comfortable measures, such as shower sand dry clothing are generally sufficient.

Coagulation:

During pregnancy, plasma fibrinogen necessary for coagulation increased as a protection against postpartum hemorrhage. As a result, the mother's body has a great ability to form clots and thus prevent excessive bleeding.

Increase in plasminogen ―necessary for lysis of clots‖ has not occurred during pregnancy. The result is that during pregnancy and the postpartum period she is at risk for thrombus (clot) formation.

Vital signs:

Temperature:

The temperature is slightly elevated: 0.5 degree for the first 24 hours and up to 38degrees is known. This rise in temperature is due to the absorption of waste products of muscular contractions of labor "dehydration".

Transient rise in temperature later is due to:

Milk engorgement ―by the 3rd - 4th day postpartum‖

Constipation.

Nervous excitation.

Pulse:

The heart rate often decreases to rate 60 to 70 b/m and is known as physiological bradycardia for 24-48 hrs. After labor it is due to the rest period after labor.

The rapid pulse may be due to pain, visitors, excitement, exhaustion, the nursing infant, hemorrhage, or infection.

Respiration:

This is in usual relation with pulse and temperature. Because of a reduction in the size of the uterus and relaxation of the abdominal wall respiration is more abdominal in character.

1. Deviation from the normal may suggest pneumonia or embolism.

Blood pressure:

No change is counted, but if hypotension is present, post-partum hemorrhage maybe suspected.

If hypertension is present (over 140/90 mm Hg) postpartum toxemia may be suspected.

3} Gastrointestinal System:

Thirst is present due to the marked fluid loss through sweat and urine.

Tendency to atony of the gastrointestinal tract, with flatulence and constipation.

Constipation may be present because of Intestinal atony, Anorexia after labor, Loss of body fluids, Laxity of the abdominal wall, Hemorrhoids, perineal trauma and episiotomy, Reflex inhibition enema in labor.

4} Urinary system:

The bladder of the postpartum woman has an increased capacity and has lost some of its muscle tone. During childbirth the urethra, bladder, and tissue around the urinary meatus may become edematous and traumatized. This often results in diminish sensitivity to fluid pressure, and many mothers have no sensation of needing to void even when the bladder is distended. Urinary retention and over-distension of the bladder may cause two complications: 1- Urinary tract infection. 2. Postpartum hemorrhage.

5} Musculoskeletal system:

Muscles and joints:

During the first few days, levels of the hormone relax in gradually subside. Ligaments and cartilage of the pelvis begin to return to their pre-pregnancy position. This can cause hip or joint pain that interferes with ambulation and exercise.

Good body mechanics and correct posture are important during this time to prevent low back pain and injury to the joint.

Abdominal wall:

During pregnancy, the abdominal walls stretch to accommodate the growing of the fetus and muscle tone diminished.

Many women, expecting that the abdominal muscles will return to the pre-pregnancy condition after labor, are dismayed to find the abdominal muscles weak, soft, and Flabby.

6} Integumentary system:

Chloasma of pregnancy usually disappears at the end of pregnancy.

Skin pigmentation gradually decrease.

Hyperpigmentation of the areolae and linea nigra may not regress completely after childbirth, and it may be permanent in some women.

Danger signs during postpartum period for the mother:

- Excessive bleeding
- Foul smelling vaginal discharge
- Fever with or without chills
- Severe abdominal pain
- Excessive tiredness or breathlessness
- Swollen hands, face and legs with severe headaches or blurred vision
- Painful, engorged breasts or sore, cracked, bleeding nipples.

Danger signs during postpartum period for the baby:

- Fever (above 38°C)
- Low body temperature (below 35.5°C)
- Fast breathing (more than 60 breaths per minute)
- Severe diarrhea or constipation Colic
- Repeated vomiting esp. projectile vomiting
- Bleeding or offensive odor from the site of the cord

Postpartum Nursing Intervention

Objectives

1. To help the mother to return to her pre-pregnant physical state with the minimum possible discomforts and no residual complications.

2- To help the mother to establish and maintain a functioning mother-child relationship.

Nursing care during Postpartum Period

- Immediate care of the mother

- Late care of the mother
- Care of the newborn

A. Immediate care of the mother

Observation and recording to:

Vital signs:

Check vital signs 2 times daily "morning and evening".

A temperature of 38c or above, for two consecutive days after the first 24hrs.isconsidered an early sign of puerperal infection.

Bradycardia is a normal physiological phenomenon.

Uterus:

Palpate the uterus daily to assess firmness, level of fundus, and the rate of involution of the uterus.

It should be well contracted, central and involutes gradually at the rate of about 1cmdaily during the first seven to ten days of the puerperium.

Breast:

The breasts are examined to note their tension and consistency, and the signs of infection.

In breast-feeding mothers, the amount and flow of milk and the condition of the nipples and areola should be observed.

Perineum

Observe perineum and suture line if present, for redness, ecchymosis, and edema or gapping. Check healing and cleanliness.

During the examination haemorrids may be noted and appropriate treatment advised.

Lochia: Check lochia for color, amount, odor, consistency, and blood clots.

Urine output: The urine output is usually recorded for the first 24 hours after delivery to ensure that the woman is passing and adequate amount of urine.

Legs: The midwife examines the patient's leg for pain and edema.

Emotional state

Phases of developing maternal role.

Postpartum blues.

Rest and sleep:

Provide for sufficient periods of rest and sleep to maintain physical and mental health, as to promote lactation (8hrs nighttime sleep and 2hrs after noon-nap are needed)

Diet:

- Provide diet high in proteins and calories to restore tissues.

- A daily requirement of 3000-3500 cal/day is needed in the form of a well-balanced diet rich in 1st class proteins, calcium, iron, vitamins, thiamine, riboflavin and ascorbic acid.
- Liberal amounts of fluids are required "the daily fluid intake should be 2.5-3 liters"(e.g. milk, juice. etc"

Hygiene:

- ✓ The women should be taken shower daily.
- ✓ The vulva and perineal care include washing or swabbing with warm water and antiseptic solution, the area must be kept clean and dry and free from infection.
- ✓ The perineum must be inspected daily if there are sutures to see that healing is taking place. Non absorbable sutures are removed on the fifth or sixth day.
- ✓ Breast care should be done before and after feeding. The nurse teaches the mother the technique of breast care and encourages her to initiate breast-feeding.

Breast-feeding:

Advantages of Breast Feeding

For baby:

- Immunological properties help prevent infections. Provides nutritional needs.
- Easily digested.
- Less sodium and protein than in cow's milk; puts less stress on newborn's kidneys.
- Calcium is better absorbed.
- Least allergenic food for infant.
- Promotes development of facial muscles, jaw, and teeth.
- Less likely to be overfed, less obesity.
- Have natural laxative effects.
- Fulfilling psychological needs.

For the mother:

- Oxytocin release aids uterine involution.
- Strong mother-infant relationship.
- Convenient; always available; no preparation.
- Cost effective.

- Less incidence of cancer breast.
- Natural contraception.

Postpartum Recovery Guideline:

Tips for Encouraging Mother Recovery during First 2 weeks:

- If mother had a vaginal delivery, tell her to use an ice pack or frozen pads with witch hazel on perineum. Use a spray bottle of warm water during or after peeing.
- Take analgesic at regular intervals. Pain begets pain, so do your best to get ahead of it.
- Take a stool softener and drink lots of water. Many hospitals won't let mother leave unless she has pooped, so make it a little easier on mother.
- Again, for C-section moms: it is a must to keep your incision clean and dry. Give it fresh air after a shower, pat it down with a towel.
- It's very important to take your temperature 2 to 4 times a day for the first 72 hours to catch a uterine or kidney infection quickly.

Tips for Encouraging Mother Recovery after 6 weeks:

- Although this is technically when the postpartum period ends, many women don't feel remotely like themselves for a full year, so be gentle with yourself.
- If you're ready to resume exercising, start slowly.
- Same holds true for sexual activity: Just because you're cleared doesn't mean that you feel ready. Listen to your body above all else. Very few mothers' experience painless sex this early after giving birth.
- The exhaustion at this point can be overwhelming. Nap as often as possible.

Tips for Encouraging Mother Recovery after 6 weeks:

- Exercise is very important at this stage, for both mental and physical health.
- Mother can do abdominal strengthening exercises in earnest, which should help alleviate some back pain.
- **Proper positioning:**
- During the first 8hrs.after labor, the mother is allowed to sleep in any comfortable position.
- After that, prone position or either lateral positions should be encouraged to facilitate involution and to help drainage of lochia.

- Sitting position is also recommended since it promotes contraction of the abdominal muscles, aids pelvic circulation, and helps drainage of lochia.
- Knee-chest position is indicated in certain conditions because it prevents RVF of the uterus and hastens its involution.
- Both supine and semi-sitting positions should be avoided.
- **Ambulation:**
- Encourage early ambulation to prevent blood stasis and deep venous thrombosis.
- Heavy activities are avoided to prevent complications.

Promote bladder and bowel function:

Bladder

- Voiding should be encouraged within 6-8hrs after labor.
- If no urine is passed after 12 hrs. Initiate simple nursing measure to induce voiding.
- If failed, catheterization, under complete a septic technique is performed.

Bowel

- There may be no bowel action for a couple of days because the bowel has probably been emptied during labor.
- Glycerin suppository may be used to relieve constipation.

Post-natal exercise: Important of exercises:

- Promotes circulation.
- Lessen the possibility of DVT and restore the muscle tone of the abdominal wall and pelvic floor.
- Give the patient a sense of wellbeing.
- Deep breathing and free movements in the bed started at the day of the delivery.
- Pelvic floor tone. Several exercises may be performed e.g. Kegel exercises

B. Late care of the mother

Health education and counseling

- The midwife nurse plays an important role as health educator and counseling which should provide the woman health education and counseling about:

- Breast feeding, definite, technique and position.
- Post-natal exercise, hygiene, rest, sleep and nutrition.
- The care of the baby which includes hygiene, prevention of infection, feeding and giving him love and sense of security feel her about the advantages of rooming in.
- Family planning methods for spacing of pregnancy.
- Stress the importance of postpartum examination. Visits and follow up to assess involution, general health and wellbeing of the mother before discharge.

C. Care of the newborn infant:

- Observing the general condition.
- Observe Temperature, skin color, newborn rash, jaundice
- Cleanliness, handling, clothing, cord care, feeding, bonding, diapering, circumcision of male infant, immunization, weaning.
- Stool & voiding [6 or more voids/day]
- Ensure adequate hours of sleep.
- Protect from environmental hazards.
- Encourage early skin to skin contact, bonding and attachment.